

7. A great deal of scientific research into global warming has happened over the last 30 years.

More recently, two new contrasting theories linked to global warming have been widely reported. These are outlined below.

Solar activity	Global dimming
A small number of scientists believe that changes in solar activity (changes in the brightness and warmth of the Sun) is causing global warming. Solar activity occurs on the surface of the Sun when nuclear reactions cause flares to be released, high winds to occur and a release of high energy particles from the Sun.	Some scientists believe that global warming is increasing the Earth's mean temperature, but that global dimming is preventing it from increasing even more. Global dimming is the reduction in the amount of radiation reaching the Earth's surface from the Sun. It is thought to be caused by an increase of aerosol particles in the Earth's atmosphere caused by pollution, dust and smog, as well as volcanic eruptions. These particles absorb solar energy and reflect sunlight back into space.

Figure 1 shows the mean atmospheric temperature, carbon dioxide concentration in the atmosphere and solar activity between 1930 and 2010.

Figure 1

Year	Mean atmospheric temperature (°C)	CO ₂ concentration (ppm)	Solar activity (arbitrary units)
1930	13.6	309	3.9
1940	13.6	311	4.0
1950	13.8	316	4.2
1960	13.9	320	4.2
1970	14.0	331	4.0
1980	14.2	342	3.9
1990	14.3	358	3.7
2000	14.4	370	3.7
2010	14.6	383	3.6

ppm = parts per million

